

Lab Experiment

Experiment 3 : Water Jug Problem using Python

Aim

To solve the Water Jug Problem using Python by finding a sequence of steps to measure the required amount of water.

Algorithm

1. Start with both jugs empty.
2. Fill the 4-liter jug.
3. Pour water into the 3-liter jug.
4. Empty the 3-liter jug when it becomes full.
5. Transfer the remaining water.
6. Repeat until 2 liters remain in the 4-liter jug.
7. Display each step.

Pseudo Code

START

Initialize Jug1 = 0 and Jug2 = 0

Fill Jug1

Pour Jug1 into Jug2

Empty Jug2

Pour remaining water from Jug1 to Jug2

Fill Jug1

Pour Jug1 into Jug2 until Jug2 is full

Print each step

STOP

Python Code

```
def water_jug(x, y, target):
```

```
    a, b = 0, 0
```

```
    while (a != target and b != target):
```

```

if a == 0:
    a = x
    print(a, b)
elif b == y:
    b = 0
    print(a, b)
else:
    pour = min(a, y - b)
    a -= pour
    b += pour
    print(a, b)

print("Target achieved:", a, b)

water_jug(4, 3, 2)

```

Output

Water Jug Problem Solution

```

C:\Users\B DEEPTI\Downloads\AI programme > python water_jug.py
1 def water_jug(x, y, target):
    a = 0
    b = 0
    while a != target and b != target:
        if a == 0:
            a = x
            print(a, b)
        elif b == y:
            b = 0
            print(a, b)
        else:
            pour = min(a, y - b)
            a -= pour
            b += pour
            print(a, b)
    print("Target achieved:", a, b)

water_jug(4, 3, 2)

```

Python Debug Console

```

PS C:\Users\B DEEPTI\Downloads\AI programme> & 'c:\Users\B DEEPTI\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\B DEEPTI\vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\libs\debugpy\launcher' '53336' '-' 'c:\Users\B DEEPTI\Downloads\AI programme\water_jug_problem.py'
4 0
3 1
0 1
4 1
2 3
Target achieved: 2 3
PS C:\Users\B DEEPTI\Downloads\AI programme>

```

Goal Achieved: 2 liters in the 4L jug.

Result

The Water Jug Problem was successfully solved using Python. The required **2 liters** of water was obtained in the **4-liter jug** by following the sequence of operations.

Experiment 4 : Cryptarithmic Problem using Python

Aim

To solve a Cryptarithmic problem using Python by assigning unique digits to letters so that the arithmetic equation is satisfied.

Algorithm

1. Import the itertools module.
2. Generate all possible digit assignments for the letters.
3. Ensure each letter has a unique digit.
4. Check that leading letters are not assigned zero.
5. Verify whether the arithmetic equation is satisfied.
6. If a valid solution is found, display it and stop.

Pseudo Code

START

Import permutations from itertools

FOR each possible digit assignment

 Assign digits to letters

 IF leading digit is 0

 Skip

 ENDIF

 IF $SEND + MORE = MONEY$

 Print solution

 STOP

 ENDIF

ENDFOR

END

Python Code

```
import itertools

def solve_cryptarithmic():
    letters = 'S E N D M O R Y'.split()

    for perm in itertools.permutations('0123456789', len(letters)):
        mapping = dict(zip(letters, perm))

        if mapping['S'] == '0' or mapping['M'] == '0':
            continue

        SEND = int(mapping['S'] + mapping['E'] + mapping['N'] + mapping['D'])
        MORE = int(mapping['M'] + mapping['O'] + mapping['R'] + mapping['E'])
        MONEY = int(mapping['M'] + mapping['O'] + mapping['N'] + mapping['E'] +
mapping['Y'])

        if SEND + MORE == MONEY:
            return mapping, SEND, MORE, MONEY

    return None

res = solve_cryptarithmic()

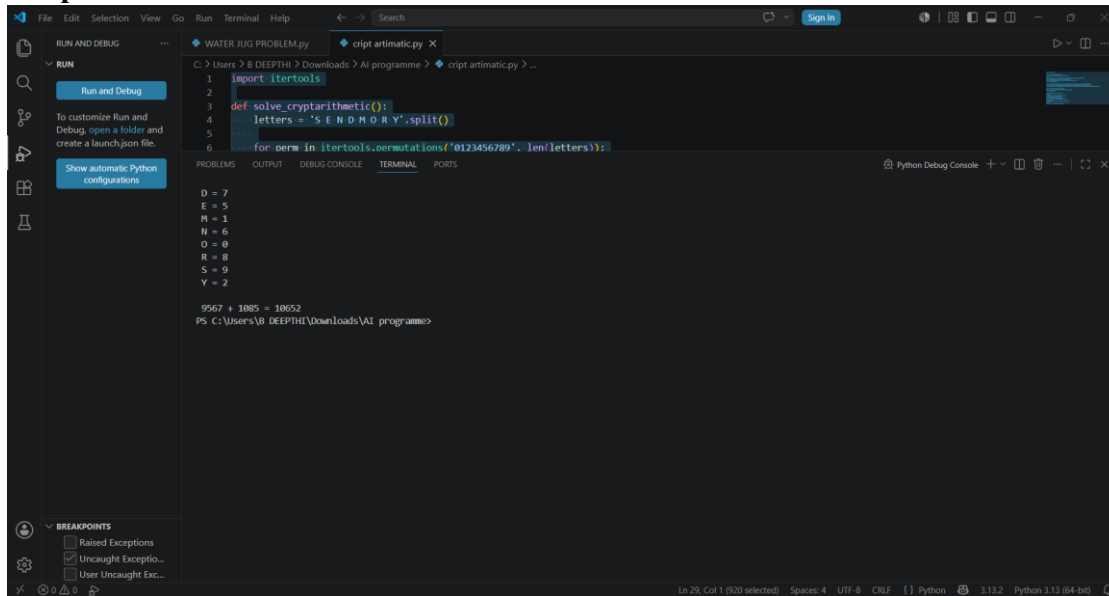
if res:
    mapping, SEND, MORE, MONEY = res
    print("Solution found:")

    for k in sorted(mapping):
        print(f" {k} = {mapping[k]}")

    print()

    print(f" {SEND} + {MORE} = {MONEY}")
else:
    print("No solution found.")
```

Output



The screenshot shows a Python IDE with a file named `cryptarithmic.py` open. The code defines a function `solve_cryptarithmic()` that uses `itertools.permutations` to solve the cryptarithm `SEND + MORE = MONEY`. The code is as follows:

```
1 import itertools
2
3 def solve_cryptarithmic():
4     letters = 'SENDMOREY'.split()
5
6     for perm in itertools.permutations('0123456789', len(letters)):
```

The output of the program is displayed in the terminal window, showing the solution for each letter and the final equation:

```
D = 7
E = 5
M = 1
N = 6
O = 0
R = 8
S = 9
Y = 2

9567 + 1085 = 10652
PS C:\Users\B DEEPTI\Downloads\AI programme>
```

Result

The Cryptarithmic problem **SEND + MORE = MONEY** was successfully solved using Python. Each letter was assigned a unique digit, and the equation was satisfied correctly.